

Table 1
Summary of the 47 data features derived from multiple data sources.

Sentinel-1 Radar Data (via Google Earth Engine)			
Index	Name	Data Source	Resolution
S1_VV	VV (Vertical–Vertical)	Sentinel-1	30 m
S1_VH	VH (Vertical–Horizontal)	Sentinel-1	30 m
S1_VV/VH	VV/VH Ratio	Sentinel-1	30 m
Sentinel-2 Spectral Bands (via Google Earth Engine)			
Predictors			
Index	Name	Data Source	Resolution
B1–B4	Coastal, Blue, Green, Red	Sentinel-2	30 m
B5–B7	Red Edge 1–3 (VNIR)	Sentinel-2	30 m
B8–B8a	NIR, Red Edge 4	Sentinel-2	30 m
B9–B12	Water vapor, Cirrus, SWIR 1–2	Sentinel-2	30 m
QA60	Cloud Mask	Sentinel-2	30 m
Spectral Indices (via Google Earth Engine)			
NDVI	Normalized Difference Vegetation Index	Sentinel-2	30 m
EVI	Enhanced Vegetation Index	Sentinel-2	30 m
Sentinel-3 Ocean & Atmosphere Bands (via Google Earth Engine)			
Oa01–Oa21	Aerosol, Chlorophyll, Water vapor	Sentinel-3	30 m
Topographic Data (via Google Earth Engine)			
Elevation	Elevation	Carto DEM	30 m
TWI	Topographic Wetness Index	Carto DEM	30 m
Slope	Slope	Carto DEM	30 m
Curvature	Plane Curvature	Derived	30 m
Aspect	Aspect	Derived	30 m
Meteorological Data			
Precipitation	Mean Annual Precipitation	CRU (UEA, 2019)	1000 m
Temperature	Mean Annual Temperature	CRU (UEA, 2019)	1000 m
Geographical Coordinates (only for non-graph methods)			
Latitude	Latitude of SOC sampling locations	Sampling coordinates	N/A
Longitude	Longitude of SOC sampling locations	Sampling coordinates	N/A

study. In summary, this multimodal dataset comprises 143 data points characterized by 47 predictor variables, as detailed in Table 1. These data are expected to provide key information for the accurate analysis and estimation of SOC levels in the Beja district, thereby supporting the objectives of the present study.

Learning from limited and heterogeneous data is a crucial capability in domains where data acquisition is costly or logistically constrained. Under these conditions, models must derive robust and generalizable patterns from sparse, noisy, and diverse information sources. This setting poses a demanding benchmark for any learning approach, revealing its capacity to develop data-efficient, resilient, and context-aware intelligence.

For this study, satellite images were obtained through the Google Earth Engine Platform.³ These images provided average spectral data corresponding to the annual recordings from the Sentinel-1, Sentinel-2, and Sentinel-3 satellites. Additionally, topographic characteristics were accessed using data from the Copernicus DEM satellite.⁴ Furthermore, satellite images related to the annual average precipitation and temperature were used, sourced from the Climatic Research Unit (CRU) databases for the year 2018.⁵ Climate variables derived from CRU were used at their native spatial support (approximately 1 km) and provide regional-scale context, whereas terrain and remote-sensing predictors

capture fine-scale spatial variability at 30 m resolution. For non-graph-based methods, two additional features—latitude and longitude of SOC sampling locations—were included to capture geographical spatial patterns in the dataset. All bands of these images were used as predictor variables to estimate SOC. Fig. 2 illustrates practical examples of visualizing Sentinel-1, -2, -3 bands.

Next subsections describe the data sources of the SOC predictors listed in Table 1, maintaining alignment between predictors and data modalities. The satellite image data used in this study have varying native spatial resolutions, as shown in Table 1. For example, Sentinel-2 spectral bands (B1–B12) are provided at 10 m, 20 m, or 60 m resolution depending on the band, with the QA60 cloud mask originally at 60 m. For consistency in the analysis, bands with coarser resolutions were optionally resampled to 30 m where appropriate, using interpolation methods to better align with higher-resolution layers. Sentinel-3 OLCI bands (Oa01–Oa21) have an original spatial resolution of approximately 300 m and were not directly acquired at 30 m; when included in analyses requiring a finer spatial scale, these bands were either used at their native resolution or resampled as needed. This approach balances spatial detail preservation and data integration requirements, avoiding unnecessary resampling when not required for the specific analysis.

This dataset poses significant challenges for machine learning models. The small sample size limits generalization and increases the risk of overfitting, while the relatively high dimensionality complicates effective feature-space exploration. Moreover, the strong heterogeneity among variables, stemming from diverse geospatial, environmental, and contextual sources, introduces complex, non-linear relationships that conventional models struggle to capture. Together, these factors make robust learning and inference particularly difficult, yet they also

³ Google Earth Engine, available at <https://earthengine.google.com/>.

⁴ Copernicus Digital Elevation Model (DEM): official website with data access and documentation.

⁵ Climatic Research Unit (CRU) databases: official website with climate data and documentation.

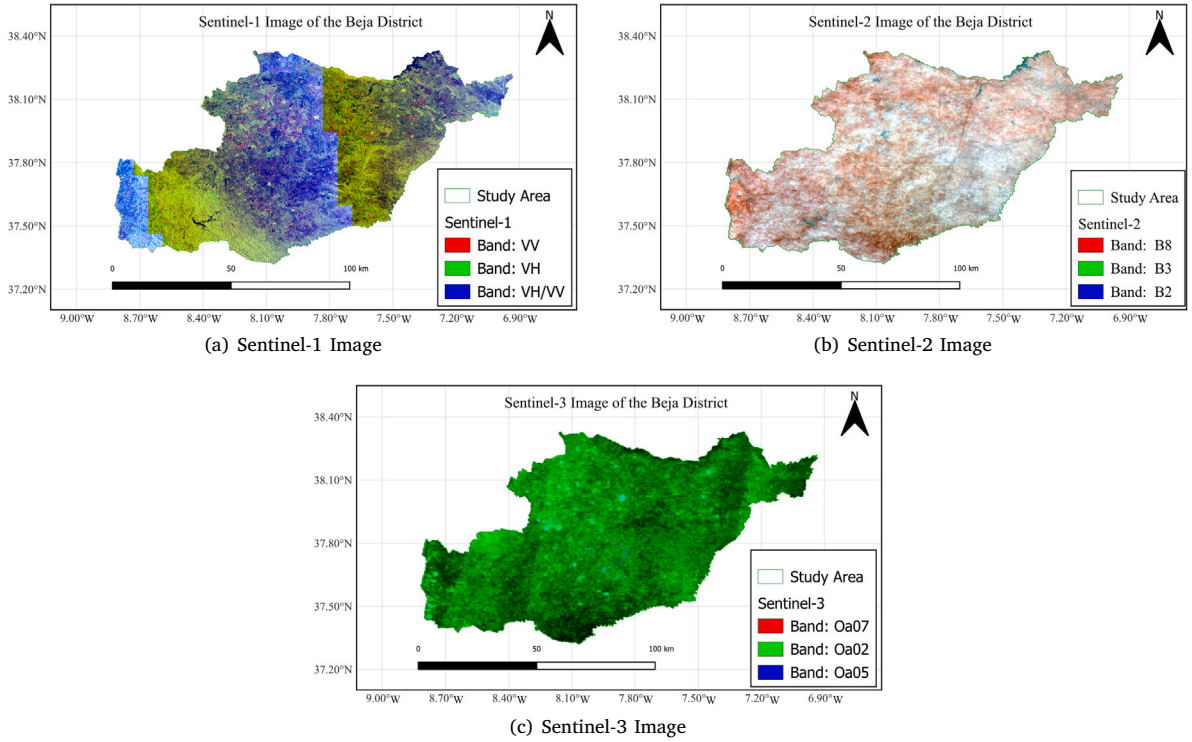


Fig. 2. Representative examples of spectral bands from the Sentinel-1, Sentinel-2, and Sentinel-3 satellites.

provide a valuable opportunity to assess the ability of GNNs to extract meaningful patterns and robust representations under data-scarce and multimodal conditions.

2.2.1. SOC data

The SOC data used in this study were obtained from two reputable sources: the European Soil Data Centre (ESDAC)⁶ and the International Soil Reference and Information Centre (ISRIC)⁷ for the year 2018. SOC data consist of soil samples collected and analyzed according to the standards of the International Soil Organization (ISO). Each soil sample is georeferenced with precise latitude and longitude coordinates and includes information on soil profile depth, specifically SOC measurements within the 0–40 cm depth range. A total of 143 georeferenced soil samples were collected within the Beja district, providing detailed information on organic carbon content and their exact spatial locations as shown in Fig. 3.

The geospatial coordinates of these samples act as a pivotal link, enabling the extraction and integration of diverse environmental and spatial features from the sources listed in Table 1, while ensuring their spatio-temporal alignment with SOC measurements. Although the satellite imagery covers a large area of thousands of square kilometers, each coordinate allows the identification of the exact pixel corresponding to the SOC sampling location.

To examine the distribution of SOC values, Fig. 4(a) presents a histogram showing right skewness, with a concentration of lower values and a long tail toward higher values (outliers). To address this issue and improve the data's suitability for predictive modeling, a natural logarithmic transformation ($\text{LN}(\text{SOC})$) was applied. This transformation effectively reduces the skewness, leading to a close-to-normal distribution, as shown in Fig. 4(b), which helps mitigate the influence of outliers and stabilize variance.

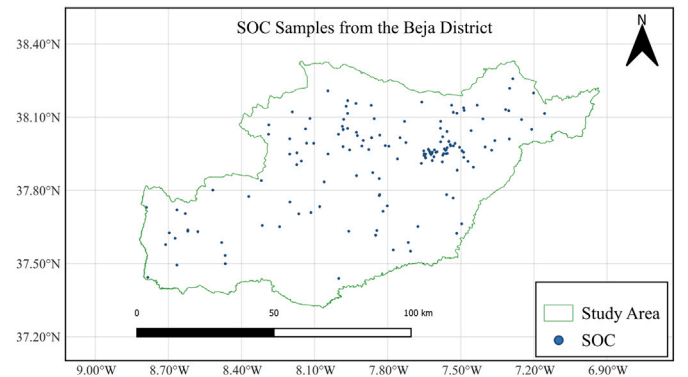


Fig. 3. Spatial distributions of surface SOC measurements.

2.2.2. Global and local spatial autocorrelation of SOC

Global spatial autocorrelation of SOC was assessed using Moran's I statistic to evaluate whether SOC values exhibit systematic spatial dependence across the study area. Moran's I measures the degree of similarity among observations as a function of spatial proximity and is defined as

$$I = \frac{n}{S_0} \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad (1)$$

where n denotes the number of sampling locations, x_i and x_j are SOC values at locations i and j , respectively, $\bar{x}$ is the mean SOC value, w_{ij} represents the spatial weight between locations i and j , and S_0 is the sum of all spatial weights.

Prior to computing Moran's I , SOC values were standardized to zero mean and unit variance. This standardization ensures that the statistic reflects relative deviations from the global mean rather than absolute magnitudes, facilitates the interpretation of the Moran scatter plot, and allows the spatial lag to be expressed on a comparable scale.

⁶ European Soil Data Centre (ESDAC): official website with soil data and related resources.

⁷ International Soil Reference and Information Centre (ISRIC): official website with global soil data and information services.

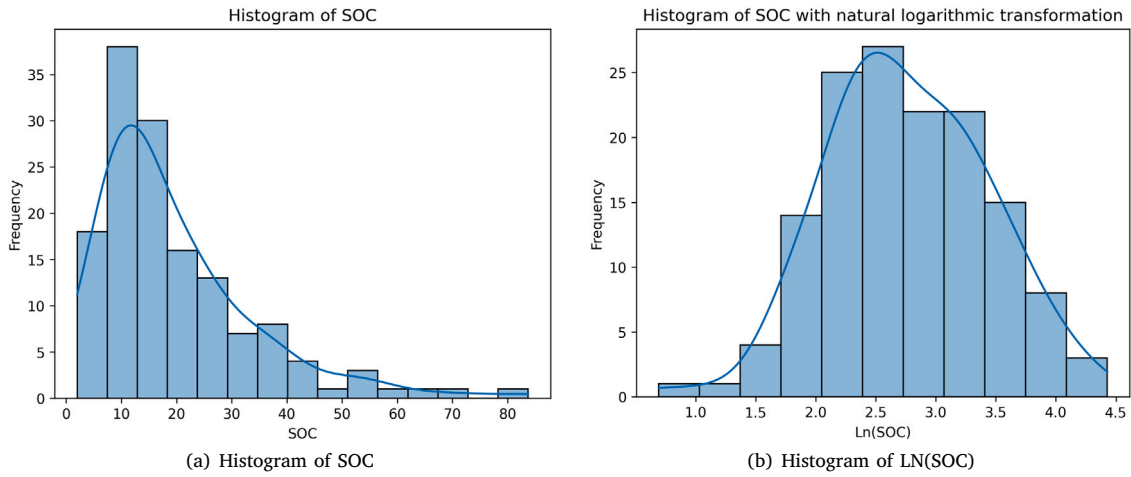


Fig. 4. Distribution of SOC data: left, the original distribution; right, the approximately normal distribution obtained after log normalization.

Consequently, the slope of the regression line in the Moran scatter plot corresponds directly to the global Moran's I coefficient.

The spatial weights matrix was constructed using a k -nearest neighbors approach ($k = 3$) based on Haversine distances between sampling locations. Weights were defined as the inverse of the inter-point distance and row-standardized so that the spatial lag corresponds to a weighted local average of neighboring SOC values.

Statistical significance was evaluated using a Monte Carlo permutation test with 999 random permutations of the observed SOC values. Under the null hypothesis of spatial randomness, SOC values are assumed to be spatially independent conditional on the specified spatial weights matrix. The observed global Moran's I value was $I = -0.065$, with a permutation-based p -value of $p = 0.179$. Therefore, the null hypothesis could not be rejected at the $\alpha = 0.05$ significance level, indicating no statistically significant global spatial autocorrelation of SOC.

Fig. 5 presents the Moran scatter plot, which provides a complementary graphical interpretation of this result. The near-zero and slightly negative slope of the regression line indicates a very weak inverse association between SOC values and the average SOC of neighboring locations. The wide dispersion of points across the four quadrants further suggests that local deviations from the global mean are not systematically aligned with their spatial context, consistent with a largely random spatial structure at the analyzed scale.

To further investigate potential localized spatial patterns that may not be captured by the global statistic, Local Indicators of Spatial Association (LISA) were examined through a Moran quadrant map. LISA decomposes the global Moran's I into location-specific contributions, allowing the identification of local clusters (high-high and low-low) and spatial outliers (high-low and low-high). This local perspective is particularly relevant in heterogeneous environmental systems, where spatial dependence may manifest only in restricted areas rather than at the regional scale.

Formally, the Local Moran's I statistic for observation i can be expressed as

$$I_i = z_i \sum_{j=1}^n w_{ij} z_j, \quad (2)$$

where z_i denotes the standardized SOC value at location i , z_j is the standardized SOC value at a neighboring location j , and w_{ij} represents the corresponding spatial weight. The local statistics I_i quantify the contribution of each observation to the global Moran's I and allow the identification of local clusters and spatial outliers by jointly considering the sign of z_i and its spatial lag.

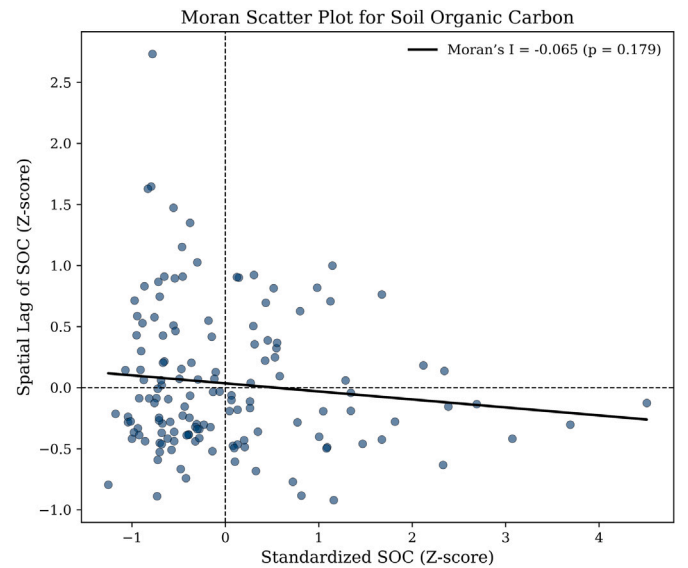


Fig. 5. Moran scatter plot for SOC showing standardized SOC values versus their spatial lag. The slope of the fitted regression line corresponds to the global Moran's I statistic.

In Fig. 6 the LISA quadrant map reveals a spatially interspersed distribution of quadrant types, with no clearly delineated or spatially coherent clusters dominated by high-high or low-low associations. Instead, high and low SOC values frequently occur adjacent to contrasting neighbors, resulting in a substantial presence of spatial outliers. This pattern is indicative of pronounced local-scale heterogeneity and suggests that SOC variability is primarily governed by site-specific conditions rather than by smooth spatial gradients.

Taken together, the global Moran's I statistic, the Moran scatter plot, and the LISA quadrant map consistently indicate that SOC in the study area does not exhibit meaningful spatial autocorrelation at the examined scale, with variability largely characterized by localized patterns rather than by broad-scale spatial dependence.

2.2.3. Satellite image data

2.2.3.1. Sentinel-1 satellite imagery.

The Sentinel-1 satellites are part of the European Union's Copernicus Programme, operated by the European Space Agency (ESA). They are equipped with Synthetic Aperture Radar (SAR), an active remote